Full Teaching Notes

Session 2 — Full Teaching Notes (comprehensive per-slide guide)

The comprehensive per-slide instructor guide — English, LTR

Course context

- **Session 2 of 32.** 60 min teaching + 60 min problem-solving (separate meeting).
- **Goal (student language):** "By the end of today, YOU will be able to read, draw, and hand-build a decision tree."
- **Before class — verify all of this:**
  - notebook `s02_notebook.ipynb` runs top-to-bottom (PART 0 demo cell is the hook's machine beat)
  - `assets/animals_table.pdf` printed — one copy per student (the paper lab depends on it)
  - blank paper for every student; pencils
  - the judge passes on the solution notebook (3/3)
- **Session timing map:** hook 0–5 · anatomy 5–15 · reading trees 15–25 · depth knob 25–35 · paper lab 35–50 · quiz 50–55 · handout 55–60.
- **Board plan:** the hook game's question path stays up ALL session (Segment A names its parts on it); vocabulary column on the right (node, branch, leaf, root, depth — never erased); the question counter from the game stays visible.

Slide 1 — Title: ماشینِ ۲۰ سؤالی (lead)

Welcome them back like a show is about to continue, because it is.

Say: "Last time you watched a machine learn. Today you become the machine — and then you beat it at its own game, on paper."

Do: notebook open at PART 0; animals_table.pdf stack ready; check every student can see the board (the hook is a BOARD game, not a screen game).

Slide 2 — بازیِ ۲۰ سؤالی (hook, beat 1)

[0:00] The human game. You think of a **wolf**. The class asks only yes/no questions. Count the questions on the board, aloud, in big numbers — the counter is a prop for slide 3.

Rules to state: any yes/no question about the animal; no letter-guessing (keep it in question form).

Steer if needed: if the class flails, hint with structure — "is it heavier than 10 kilos?" Good groups naturally ask weight and ear questions; both will be used later.

When they get it: do NOT clear the board. The question path drawn as it happened is slide 3's raw material.

Fallback: this beat needs nothing but a board. Never skip it.

Slide 3 — شما همین الان یک درخت بودید (hook, beat 2)

[~3:00] The reveal. Redraw the game's path on the board as a proper diagram: circle for the first question, two arrows out, circle for the second, arrows into the verdict "گرگ".

Say (verbatim): "Every answer was a door. Every door opened a new path. You just walked a tree — a decision tree. And the machine from last session? It plays exactly this game. Its red line was ONE question. One circle, two doors."
Expect: "Wait — the red line was a question?!" — that connection is the whole point. Let a STUDENT say it if possible.

Slide 4 — حالا ماشین بازی کند (hook, beat 3)

[RUN: PART 0 demo cell] — one cell, both trees side by side: depth 1 (the puppy → گربه, last session's failure) and depth 2 (puppy → سگ کوچک, fixed).

Say: "Last session the machine was allowed ONE question and got the puppy wrong. Watch what happens when it's allowed TWO."

Expect: "It fixed it?!" Then close the hook: **"You've been playing decision trees your whole life. Today you learn to draw them — on paper and in code."**

Fallback if the cell fails: slides 7 and 9 carry the same tree pictures — narrate from them. Never skip the hook's story.

Slide 5 — کالبدشکافی درخت (definition)

[5:00] Segment A. The NAME slide — the experience (the game) already happened, so this lands warm.

Do: point at the BOARD path while reading each definition — node = question circle, branch = arrow door, leaf = verdict room. The slide's vocabulary must be written into the board's vocabulary column and never erased.

[MISCONCEPTION]: the tree does NOT think or adapt mid-game. The whole path is fixed in advance; every animal walks the same corridors. Say it explicitly: "A tree cannot change its mind. The wolf and the elephant walk the same hallways."

CHECK (cold call): "How many leaves can a 2-question tree have, at most?" (4 — 2×2. This tree has 3.)

Slide 6 — ماشینِ جلسهٔ ۱ = یک سؤال

[RUN: notebook cell A1] — the depth-1 tree picture. Visually identical mechanic to last session's red line.

Say: "Last session's line and this picture are the SAME thing — a single question with two doors. Only the drawing changed."

Why this slide exists: it seals the continuity. Session 1's model was a tree all along; today they learn its language.

Slide 7 — ریشه: اولین سؤال (definition)

[15:00] Segment B. NAME: **ریشه (root)** — the first fork every animal meets.

[RUN: notebook cell A2] — the depth-2 `plot_tree`. Read the tree aloud together, door by door, for a wolf: weight ≥ 14? yes → big-dog room — ONE question only. The heavy side was already sorted; the tree stopped asking.

Interaction: compare with the class's own first questions from the hook game. Most asked weight first too — but with different thresholds (7, 10, 20). Park that observation: "the machine picked exactly 14. Why 14 and not 7? Next session." (This is the cliffhanger's seed — plant it here, water it at the end.)

[MISCONCEPTION]: the tree does NOT give every animal the same number of questions. The big dogs get one; cats and small dogs get two. A tree asks only where animals are still mixed — and WHY it stops there is exactly the next session's story.

Slide 8 — دو سؤال، سه اتاق (wow)

The punchline picture — the machine's depth-2 tree (the slide twin of the notebook's `plot_tree`: same tree, same thresholds, `random_state=42`; the notebook keeps the authentic matplotlib output).

Say: "One question split light from heavy. The heavy side was DONE — all big dogs, room right there. Only the light side needed a second question — ears — to split cats from small dogs. The tree asks where animals are still mixed, and stops where they aren't."

ASK before running cell A3: "Which room does the puppy reach now?" — they already know from PART 0; let them say it out loud for the satisfaction.

Slide 9 — فیل کدام در را می‌گذرد؟ (quiz)

**[~22:00 — placed AFTER the depth-2 tree picture, by design: the quiz needs it.]

INTERACTION (2 min). The elephant (3000 kg, ears 20): the class shouts the door at the node on the projected depth-2 tree that the previous slide just revealed.

Trace: `وزن ≤ ۱۴؟` yes → and the big-dog room is RIGHT THERE. On the heavy side the tree asks no second question — the room is the very next door.

Say: "The elephant walks the same corridors as everyone — and gets its verdict after a single question. A leaf says ONE word for whatever reaches it — even a 3-ton elephant gets called a big dog."

Honest point (say it plainly): the ear question lives on the LIGHT side only. The tree does not give every animal the same number of questions — it stops asking where the animals are already sorted. (Do not name "purity" yet — that is Session 3's word; today it is just "already sorted".)

Slide 10 — مسیر را بخوان: توله‌سگ (quiz)

[RUN: notebook cell A3] — visitors: wolf, puppy, elephant. Students trace each path BEFORE the machine answers.

Trace the wolf first: `وزن ≤ ۱۴؟` yes → big-dog room — one question, done. **Then the puppy:** `وزن ≤ ۱۴؟` no → light corridor. `گوش ≤ ۸.۲۵؟` yes → small-dog room. Correct!

Say: "The second question — ears — is what separates small dogs from cats. Last session's mistake had a cure: one more question."

USE-IT landed here: every student has now traced a path and predicted before checking. That is "reading a tree."

Slide 11 — پیچِ عمق (definition)

[25:00] Segment C. NAME: **عمق (depth)** — the longest question path. `max_depth` = the knob that limits it.

[RUN: notebook cell A4] — depth 1 vs depth 2 comparison with accuracy printed: 73% → 100%.

[MISCONCEPTION]: more depth is NOT automatically smarter. The machine has no judgment about when to stop — WE turn the knob. The honest full story (overfitting) arrives in Block 2; do not teach it now, but do not lie about it either — "more questions is not always better; WHY is a story for later" is the exact scripted line.

CHECK: "Depth 1: how many leaves? Depth 2? Depth 3, at most?" (2 / 4 / 8)

Slide 12 — دستت را روی پیچ بگذار (codespot)

[RUN: notebook cell A5 — students change the number THEMSELVES and re-run]

The race: first student to find the smallest perfect depth (2) shouts it. Then run `max_depth=10` together — the tree becomes an unreadable jungle; let them laugh at it.

Say: "More questions, finer rooms — but would YOU trust a tree that asks 10 questions about 11 animals? Hold that feeling. Block 2 will name it."

Slide 13 — آزمایشگاه: درختِ خودت را بکش (hook — the Lab)

[35:00] THE LAB — this session's Ship. Hand out `animals_table.pdf` + blank paper.

Rules (state exactly): only yes/no questions about the table's two numbers (weight, ear length). Goal: every animal reaches the right room (cat / big dog / small dog). Any tree that works, wins.

Circulate. Stuck student hint: "which question would split the cats away from the dogs first?"

Timing: ~10 minutes drawing. Do not rush this — it is the emotional core of the session.

[MISCONCEPTION] (say it during circulation): there is no single correct tree. Two students with different first thresholds both have valid trees.

Slide 14 — نبردِ درخت‌ها (quiz)

[~45:00] The battle. Pairs swap papers. Test each other's trees on the surprise zoo: rabbit, fox, hamster, sparrow, lion (the cell C1 list — read them out).

Protocol: verdicts read door-by-door, out loud. **The funniest failure gets applause** (a sparrow called a cat; a rabbit called a small dog). Each failure is diagnosed with one question: "WHICH question let you down?"

Verified verdicts to expect (depth-2 machine tree): rabbit → سگ کوچک, fox → گربه, hamster → گربه, sparrow → گربه, lion → سگ بزرگ. Student trees will vary — that's the point.

Slide 15 — درختِ تو = درختِ ماشین (dark)

[~48:00] Connect. [RUN: notebook cell A6] — the machine's depth-2 tree. If the room allows, photograph one student's paper tree and project it next to the machine's.

Say: "Same game. Same shape. Yours is on paper; the machine's is in memory. And yours — you can tape it to a wall."

Ship logistics (show on screen): photo the tree → `s02_tree.jpg` → commit to their repo tonight.

Slide 16 — امروز یاد گرفتی (quiz)

[50:00] CONSOLIDATION — cold call, one question per student:

1. Parts of a tree? (node/question, branch/door, leaf/room)
2. Why did depth-2 fix the puppy? (the ear question separates small dogs from cats)
3. What is max_depth? (the most questions the machine may ask)
4. Can two different trees both be correct? (yes — many valid trees)

If anyone stumbles on Q1–3, re-explain NOW, not at home.

Slide 17 — تحویل امشب (handout)

[55:00] The Ship + the loop, said OUT LOUD: photo of the paper tree committed tonight; three small notebook problems (20–30 min); next session opens by solving them TOGETHER from their attempts.

Show the judge command: `python judge/run.py s02_notebook.ipynb`

Slide 18 — تمرین‌های خانه (handout)

One sentence of story per problem:

- **B1** — count the tree's questions and rooms (reading a tree's shape)
- **B2** — which room does the elephant reach? (tracing a path)
- **B3** — the smallest perfect depth (the knob, measured)

Deadline: before the problem-solving session.

Slide 19 — جلسةٌ بعد (wow, cliffhanger)

[60:00 — end on time.]

Cliffhanger (verbatim): "Your paper tree asked smart questions. But WHO taught the MACHINE to pick its questions? It read the same 11 animals and chose `weight ≥ 14` — not 7, not 20. Why that one? Next time: you become the question-picker."

After class (48h): fill the retro note in the session README.